Hugon Lee

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Research Interests

Micromechanics of composites · Manufacturing process modeling · Data-driven design and optimization · Physics-informed machine learning · Agentic AI

Education

Ph.D. in Mechanical Engineering, KAIST, Rep. of Korea Feb. 2022 – Aug. 2026 (expected)
Advisor: Prof. Seunghwa Ryu

- Thesis: *Interface and interphase effects in stiff and soft composites: micromechanics and physics-informed machine learning*
- GPA (M.S.–Ph.D., cumulative): 4.04/4.30

M.S. in Mechanical Engineering, KAIST, Rep. of Korea Feb. 2020 – Feb. 2022
Advisor: Prof. Seong Su Kim

- Thesis: *(A) study on permeability prediction of fiber-reinforced composite materials based on the representative volume element*

B.S. in Mechanical Engineering, KAIST, Rep. of Korea Mar. 2014 – Feb. 2020

Awards & Honors

Outstanding Paper Award, KSME CAE & Applied Mechanics Division May 2026
Title: Multi-fidelity surrogates for efficient prediction of SFRP mechanical properties

Outstanding Paper Award, KSME CAE & Applied Mechanics Division Apr. 2025
Title: Modeling and simulation of DLP 3D printing process using Abaqus

Academic Excellence Award, Mechanical Engineering, KAIST Fall 2018 & Spring 2019
Awarded to the undergraduate students with GPA over 3.8/4.3 in the Dept. of Mech. Eng.

Dean's list, College of Engineering, KAIST Fall 2018
Awarded to the top 3% of undergraduate students in the Dept. of Mech. Eng. for outstanding GPA

Publications

† Equal contribution   * Corresponding author

13. Kim, H., **Lee, H.**, Park, M., & Ryu, S.* "Morphology-, noise-, and resolution-robust ultrasound elasticity imaging with Fourier neural operator." *Computer Methods and Programs in Biomedicine*, 283, 109432 (2026). 10.1016/j.cmpb.2026.109432

12. Lee, I., Chae, H., Park, J., Shin, J., **Lee, H.***, & Ryu, S.* "Comparative assessment of composition- and structure-based surrogate models across 2D materials databases." *Physical Chemistry Chemical Physics*, 28(16), 10119–10130 (2026). 10.1039/D5CP04814A

11. **Lee, H.**†, Moon, H.†, Lee, J.†, & Ryu, S.* "Toward knowledge-guided AI for inverse design in manufacturing: A perspective on domain, physics, and human-AI synergy." *Advanced Intelligent Discovery*, e202500107 (2025). 10.1002/aidi.202500107 (Perspective article)

10. Cho, H., **Lee, H.**, & Ryu, S.* "Sequential multi-objective Bayesian optimization of air-cooled battery thermal management system with spoiler integration." *Journal of Energy Storage*, 114, 115586 (2025). 10.1016/j.est.2025.115586

9. **Lee, H.**†, Yeo, J.†, Kong, K., Myeong, D., Jang, D., Lee, J., Choi, H., Kim, N.*, & Ryu, S.* "Bayesian optimization of tailgate rib structures enhancing structural stiffness under manufacturing constraints of injection molding." *Journal of Manufacturing Processes*, 134, 739–748 (2025). 10.1016/j.jmapro.2024.12.064

8. **Lee, H.**, Yeo, J., Yu, J., Moon, H., & Ryu, S.* "Mean-field homogenization of liquid metal elastomer composites: Comparative study and exact dilute solution of core-shell inclusion." *Journal of Physics: Condensed Matter*, 36(50), 505702 (2024). 10.1088/1361-648X/ad765e

7. **Lee, H.**†, Lee, M.†, Jung, J., Lee, I.*, & Ryu, S.* "Enhancing injection molding optimization for SFRPs through multi-fidelity data-driven approaches incorporating prior information in limited data environments." *Advanced Theory and Simulations*, 7(8), 2400130 (2024b). 10.1002/adts.202400130

6. Jo, J.†, Park, M.†, Kang, S.†, **Lee, H.**†, Gu, C.-Y., Kim, T.-S., & Ryu, S.* "Data-driven prediction of strain fields in auxetic structures and non-contact validation with mechanoluminescence for structural health monitoring." *International Journal of AI for Materials and Design*, 1(2), 48–60 (2024). 10.36922/ijamd.3539

5. Jung, J., Park, K., **Lee, H.**, Cho, B., & Ryu, S.* "Comparative study of multi-objective Bayesian optimization and NSGA-III based approaches for injection molding process." *Advanced Theory and Simulations*, 7(7), 2400135 (2024a). 10.1002/adts.202400135

4. Jo, J.†, Park, K.†, Song, H.†, **Lee, H.**, & Ryu, S.* "Innovative 3D printing of mechanoluminescent composites: Vat photopolymerization meets machine learning." *Additive Manufacturing*, 90, 104324 (2024). 10.1016/j.addma.2024.104324

3. Park, D.†, Lee, J.†, **Lee, H.**, Gu, G. X., & Ryu, S.* "Deep generative spatiotemporal learning for integrating fracture mechanics in composite materials: inverse design, discovery, and optimization." *Materials Horizons*, 11(13), 3048–3065 (2024). 10.1039/D4MH00337C

2. **Lee, H.**, Lee, S.*, & Ryu, S.* "Advancements and challenges of micromechanics-based homogenization for the short fiber reinforced composites." *Multiscale Science and Engineering*, 5(3), 133–

146 (2023). 10.1007/s42493-024-00100-2 (Review article)

1. Lee, J., Park, D., Lee, M., **Lee, H.**, Park, K., Lee, I., & Ryu, S.* "Machine learning-based inverse design methods considering data characteristics and design space size in materials design and manufacturing: a review." *Materials Horizons*, 10(12), 5436–5456 (2023). 10.1039/D3MH00039G (Review article)

Selected Oral Presentations

5. **Lee, H.**, & Ryu, S.* "Multi-fidelity surrogate for mechanical properties of short fiber reinforced plastics: leveraging information from theory to experiment." The 9th Asian Pacific Congress on Computational Mechanics / The 7th Australasian Conference on Computational Mechanics (APCOM-ACCM 2025), Brisbane, Australia (Dec. 2025).

4. **Lee, H.**, & Ryu, S.* "Micromechanical analysis on interphase effects of liquid metal elastomer composites." The 2024 Conference of the Korean Society of Mechanical Engineers (KSME 2024), Jeju, Republic of Korea (Nov. 2024).

3. **Lee, H.**†, Jo, J.†, & Ryu, S.* "Numerical implementation of digital light processing 3D printing process using Abaqus." The 26th International Congress of Theoretical and Applied Mechanics (ICTAM 2024), Daegu, Republic of Korea (Aug. 2024).

2. **Lee, H.**, Jung, J., & Ryu, S.* "Data-driven optimization of injection molding process parameters: comparisons of parts with varying complexity." The 2023 Spring Conference of the Korean Society of Mechanical Engineers (KSME), CAE & Applied Mechanics Division, Busan, Republic of Korea (May 2023).

1. **Lee, H.**, & Kim, S. S.* "A study on the optimal curing process of filament wound tow-preg laminated composites through multi-scale modeling." The 24th International Conference on Composite Structures (ICCS24), Virtual, Porto, Portugal (Jun. 2021).

Teaching & Mentoring

Academic Talk & Teaching

LLM tool-calling tutorial, KAIST Industry-Academia Open Course Feb. 2026
Lecture and hands-on tutorial on LLM usage and tool-calling for homogenization workflows

LLM tool-calling tutorial, 2026 Winter AI Class of KSCM Feb. 2026
Lecture and hands-on tutorial on LLM usage and tool-calling for homogenization workflows

Invited talk, Prof. Patrick C. Lee group, University of Toronto, Canada Nov. 2023
Introduction to multi-fidelity machine learning and its utilization for experimental data

Student Mentoring

Inhyo Lee Jan. 2025 – Oct. 2025
Research design & manuscript preparation; published in *Physical Chemistry Chemical Physics*

Sangwoo Jung Jun. 2024 – Oct. 2025
Master's research: modeling of DLP additive manufacturing process

Hanbin Cho Aug. 2023 – Aug. 2024
Master's research: numerical analysis & Bayesian optim.; published in *Journal of Energy Storage*

Hyeonseong Jo Feb. 2021 – Feb. 2022
Master's research: experimental design, permeability analysis, numerical modeling

Professional Service

Journal Reviewer: Nature Communications, Mechanical Systems and Signal Processing, Applied Composite Materials, Progress in Additive Manufacturing, Journal of Thermal Analysis and Calorimetry, Discover Mechanical Engineering

Skills

- **Programming:** Python, Fortran, MATLAB, LaTeX
- **Simulation:** Abaqus, FEniCSx, COMSOL, HyperWorks, Moldflow, ANSYS Fluent
- **Modeling:** Micromechanics/homogenization, surrogate modeling (multi-fidelity), Bayesian optimization, polymer process modeling
- **Workflow:** Script-based simulation workflows with automated pre-/post-processing, machine learning and inverse design (PyTorch, GPy), multi-agent orchestration and tool-calling (LLM), scientific visualization (Illustrator, PyVista), version control (Git)
- **Experimental:** Experimental design for material evaluation and process characterization, simulation–experiment comparison
- **Languages:** Korean (native), English (fluent)

References

Dr. Seunghwa Ryu (Ph.D. advisor)
Head, Department of AX, KAIST
Endowed Chair Professor, Department of Mechanical Engineering, KAIST

Dr. Patrick C. Lee
Assistant Professor, Dept. of Mechanical and Industrial Engineering, University of Toronto

Dr. Sangryun Lee
Associate Professor, Dept. of Mechanical and Biomedical Engineering, Ewha Womans University